

2-2-3: Examining Authentication, Authorization, and Accounting (AAA)

At the end of this episode, I will be able to:

1. Identify and explain the importance of authentication, authorization and accounting (AAA), given a scenario.

Learner Objective: *Identify and explain the importance of authentication, authorization and accounting (AAA), given a scenario*

Description: In this episode, the learner will examine a technology used to centrally manage authentication, authorization and accounting called the Remote Authentication Dial-in User Service or RADIUS.

-
- What is authentication?
 - The process of validating the identity of a user or system
 - What is authorization?
 - Granting or denying specific permissions and access rights to authenticated users or entities, controlling their actions within a system or resource.
 - What is accounting?
 - In the context of AAA, the process of tracking and recording user activities and resource usage within a system, providing an audit trail of actions for accountability, compliance, and monitoring purposes.
 - Can we bring AAA together?
 - These are brought together commonly for remote access with the Remote Authentication Dial-In User Service or RADIUS
 - RADIUS (Remote Authentication Dial-In User Service) is a networking protocol to centralize and manage user authentication, authorization, and accounting (AAA) for remote network access. (RFC 2865)
 - Where is RADIUS commonly used?
 - Network Access control: Authenticate and authorize users accessing a network, such as Wi-Fi or VPN, ensuring only authorized individuals gain access.
 - Wireless Networking: Secure access to Wi-Fi networks in enterprises and public hotspots by verifying user credentials.
 - Remote Access Servers: Control access to remote access servers like dial-up and virtual private network (VPN) servers.
 - Internet Service Providers (ISPs): Authenticate and authorize users connecting to ISP networks, managing bandwidth and services.
 - What ports does the RADIUS protocol use?
 - UDP port 1645 or 1812 for authentication
 - UDP port 1646 or 1813 for accounting.
-
- Additional Resources:
 - If applicable